



Converting fractions, decimals, and percentages

DECIMALS, FRACTIONS, AND PERCENTAGES ARE DIFFERENT WAYS OF WRITING THE SAME NUMBER.

SEE ALSO

◀ 44–45 Decimals

◀ 48–55 Fractions

◀ 60–63 Percentages

The same but different

Sometimes a number shown one way can be shown more clearly in another way. For example, if 20% is the grade required to pass an exam, this is the same as saying that $\frac{1}{5}$ of the answers in an exam need to be answered correctly to achieve a pass mark or that the minimum score for a pass is 0.2 of the total.

Changing a decimal into a percentage

To change a decimal into a percentage, multiply by 100.

$$0.75 \rightarrow 75\%$$

$$\begin{array}{ccccc} 0.75 & \times & 100 & = & 75\% \\ \text{Decimal} & & \text{Multiply by 100} & & \text{Percentage} \end{array}$$

decimal point in 0.75 moved two places to right to make 75

75%

PERCENTAGE

A percentage shows a number as a proportion of 100.

► All change

The three ways of writing the same number are shown here: decimal (0.75), fraction ($\frac{3}{4}$), and percentage (75%). They look different, but they all represent the same proportion of an amount.

Changing a percentage into a decimal

To change a percentage into a decimal, divide it by 100.

$$75\% \rightarrow 0.75$$

decimal point added two places to left of last digit

$$\begin{array}{ccccc} 75\% & \div & 100 & = & 0.75 \\ \text{Percentage} & & \text{Divide by 100} & & \text{Decimal} \end{array}$$

Changing a percentage into a fraction

To change a percentage into a fraction, write it as a fraction of 100 and then cancel it down to simplify it, if possible.

$$75\% \rightarrow \frac{75}{100}$$

divide by highest number that goes into 75 and 100

$$75\% \rightarrow \frac{75}{100}$$

Turn the percentage into the numerator of a fraction with 100 as the denominator.

$$\begin{array}{c} \div 25 \\ \div 25 \end{array}$$

$$\frac{3}{4}$$

Fraction canceled down into its lowest terms.